Differential Equations and Applications
MATH 205 & MTH 215

Higher Order Differential Equations

Higher Order DEs:

$$2y''' - 5y' - 7y = 0 \quad , \quad xy'' - (2e^{x})y' + \frac{y}{x} = 0$$

Constant Coefficients **Variable Coefficients**

$$2y'' - 5y' - 7y = 0 \quad , \quad 2y'' - 5y' - 7y = tan\ x$$

Homogeneous **Non- Homogeneous**

$$2y'' - 5y' - 7y = 0 \quad , y(0) = 1 \quad y'(0) = -2$$

Initial Value Problem (IVP)

$$2y'' - 5y' - 7y = 0 \quad , y(2) = 1 \quad y(5) = -2$$

Boundary Value Problem (BVP)

Linear Independence of Functions:

*$f(x)$ and $g(x)$ are said to be linearly dependent functions if $f(x) = c\, g(x)$ for some constant c.

Examples:

$\{3e^x, 6\, e^x\}, \{4cos2x,\ 10cos\ 2x\}, \{x^3 - 2x, -x^3 + 2x\}$

Note: The following is not dependent functions

$$\{2x, 3x^2\}\ , \{sin\ x\ , cos\ x\}\ , \{e^{2x}, e^{3x}\}$$

, they are said to be linearly independent functions

$*\ f(x)\ , g(x),$ and $h(x)$ are said to be linearly dependent functions if one of them, say, $f(x)$ can be written linearly in terms of the others

i.e. $f(x) = c_1 g(x) + c_2 h(x)$ for some 2 constant c_1, c_2

Example: $\{e^x,\ \ e^{-x}\ , sinh\ x\}$

$$sinh\ x = \frac{1}{2}\, e^x - \frac{1}{2} e^{-x}$$

How to test linear independency of functions:

$$W(f,g) = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix}$$

$$if \quad W(f,g) \neq 0 \rightarrow f, g \; are \; L.independent$$

$$W(f,g,h) = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f'' & g'' & h'' \end{vmatrix}$$

$$if \quad W(f,g,h) \neq 0 \rightarrow f, g, h \; are \; L.independent$$

You can check the above examples.

Homogeneous with constant coefficients

$$a_3 y''' + a_2 y'' + a_1 y' + a_0 y = 0$$

This DE has 3 L. independent solutions

$$y_1(x), y_2(x), y_3(x)$$

The general solution of this DE is given by

$$y_{G.S.} = c_1 y_1(x) + c_2\, y_2(x) + c_3 y_3(x)$$

, for 3 arbitrary constants c_1, c_2, c_3

The general solution of any Homogeneous ODE with order N consists of the superposition of N linear independent solutions...

If $a_n \frac{d^n y}{dx^n} + a_{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \cdots \ldots . + a_1 \frac{dy}{dx} + a_0 y = 0$

Then the solution should be

$$y_{G.S.} = c_1 y_1(x) + c_2 y_2 + \cdots . + c_n y_n(x)$$

Where $W(y_1, y_2, \ldots, y_n) \neq 0$

Example: If the two functions $y_1 = e^{3x}$, and $y_2 = e^{-3x}$ are both solutions of the homg. D.E. $y'' - 9y = 0$ and $W(e^{3x}, e^{-3x}) \neq 0$ then the general solution is given by: $y = c_1 e^{3x} + c_2 e^{-3x}$.

Example: If the two functions $y_1 = sinh3x$, and $y_2 = e^{-3x}$ are both solutions of the homg. D.E. $y'' - 9y = 0$ and $W(sinh3x, e^{-3x}) \neq 0$ then the general solution is given by: $y = c_1 sinh3x + c_2 e^{-3x}$.

Example:- If the three functions $y_1 = e^{x}$, $y_2 = e^{2x}$, and $y_3 = e^{3x}$ are solutions of the homg. D.E. $y''' - 6y'' + 11y' - 6y = 0$ and $W(e^{x}, e^{2x}, e^{3x}) \neq 0$ then the general solution is given by: $y = c_1 e^{x} + c_2 e^{2x} + c_3 e^{3x}$.

HOMOGENEOUS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

Given

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots \ldots . + a_2 y'' + a_1 y' + a_0 y = 0$$

Then, the function $\boldsymbol{y = e^{mx}}$ is a solution of the above DE under n values of m if $a_n \neq 0$

AUXILIARY EQUATION We begin by considering the special case of the second-order equation

$$ay'' + by' + cy = 0, \tag{2}$$

where a, b, and c are constants. If we try to find a solution of the form $y = e^{mx}$, then after substitution of $y' = me^{mx}$ and $y'' = m^2e^{mx}$, equation (2) becomes

$$am^2e^{mx} + bme^{mx} + ce^{mx} = 0 \qquad \text{or} \qquad e^{mx}(am^2 + bm + c) = 0.$$

As in the introduction we argue that because $e^{mx} \neq 0$ for all x, it is apparent that the only way $y = e^{mx}$ can satisfy the differential equation (2) is when m is chosen as a root of the quadratic equation

$$am^2 + bm + c = 0. \tag{3}$$

This last equation is called the **auxiliary equation** of the differential equation (2). Since the two roots of (3) are $m_1 = (-b + \sqrt{b^2 - 4ac})/2a$ and $m_2 = (-b - \sqrt{b^2 - 4ac})/2a$, there will be three forms of the general solution of (2) corresponding to the three cases:

- m_1 and m_2 real and distinct $(b^2 - 4ac > 0)$,
- m_1 and m_2 real and equal $(b^2 - 4ac = 0)$, and
- m_1 and m_2 conjugate complex numbers $(b^2 - 4ac < 0)$.

CASE I: DISTINCT REAL ROOTS Under the assumption that the auxiliary equation (3) has two unequal real roots m_1 and m_2, we find two solutions, $y_1 = e^{m_1x}$ and $y_2 = e^{m_2x}$. We see that these functions are linearly independent on $(-\infty, \infty)$ and hence form a fundamental set. It follows that the general solution of (2) on this interval is

$$y = c_1e^{m_1x} + c_2e^{m_2x}. \tag{4}$$

CASE II: REPEATED REAL ROOTS When $m_1 = m_2$, we necessarily obtain only one exponential solution, $y_1 = e^{m_1x}$. From the quadratic formula we find that $m_1 = -b/2a$ since the only way to have $m_1 = m_2$ is to have $b^2 - 4ac = 0$. It follows from (5) in Section 4.2 that a second solution of the equation is

$$y_2 = e^{m_1x}\int \frac{e^{2m_1x}}{e^{2m_1x}}\,dx = e^{m_1x}\int dx = xe^{m_1x}. \tag{5}$$

In (5) we have used the fact that $-b/a = 2m_1$. The general solution is then

$e^{\pm i\varphi} = \cos\varphi \pm i\sin\varphi$

$$y = c_1e^{m_1x} + c_2xe^{m_1x}. \tag{6}$$

CASE III: CONJUGATE COMPLEX ROOTS If m_1 and m_2 are complex, then we can write $m_1 = \alpha + i\beta$ and $m_2 = \alpha - i\beta$, where α and $\beta > 0$ are real and $i^2 = -1$. Formally, there is no difference between this case and Case I, and hence

$$y = C_1e^{(\alpha+i\beta)x} + C_2e^{(\alpha-i\beta)x}.$$

$$y = c_1e^{\alpha x}\cos\beta x + c_2e^{\alpha x}\sin\beta x = e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x). \tag{8}$$

Summary: – $ay'' + by' + cy = 0$

The auxiliary equation is given by:

$$am^2 + bm + c = 0$$

Then $m_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ ➔ Three cases :

(Distinct values, repeated values, complex values)

$$m = m_1, m_2$$
$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

$$m = m_1, m_1$$
$$y = c_1 e^{m_1 x} + c_2 x e^{m_1 x}$$

$$m = a \pm jb$$
$$y = e^{ax}(c_1 cosbx + c_2 \sin bx$$

EXAMPLE 1 Second-Order DEs

Solve the following differential equations.

(a) $2y'' - 5y' - 3y = 0$ **(b)** $y'' - 10y' + 25y = 0$ **(c)** $y'' + 4y' + 7y = 0$

SOLUTION We give the auxiliary equations, the roots, and the corresponding general solutions.

(a) $2m^2 - 5m - 3 = (2m + 1)(m - 3) = 0, \quad m_1 = -\frac{1}{2}, m_2 = 3$

From (4), $y = c_1e^{-x/2} + c_2e^{3x}$.

(b) $m^2 - 10m + 25 = (m - 5)^2 = 0, \quad m_1 = m_2 = 5$

From (6), $y = c_1e^{5x} + c_2xe^{5x}$.

(c) $m^2 + 4m + 7 = 0, \; m_1 = -2 + \sqrt{3}i, \quad m_2 = -2 - \sqrt{3}i$

From (8) with $\alpha = -2, \beta = \sqrt{3}$, $y = e^{-2x}\left(c_1 \cos \sqrt{3}x + c_2 \sin \sqrt{3}x\right)$. ■

Examples:- Solve the following DE:

$$\boldsymbol{y'' - 5y' - 6y = 0}$$

$$m^2 - 5m - 6 = 0 \qquad m = -1,6$$

$$y_{G.S.} = c_1 e^{-x} + c_2 e^{6x}$$

$$\boldsymbol{y'' + 6y' + 9y = 0}$$

$$m^2 + 6m + 9 = 0 \qquad m = -3, -3$$

$$y_{G.S.} = c_1 e^{-3x} + c_2 x e^{-3x}$$

$$\boldsymbol{y'' - 6y' + 13y = 0}$$

$$m^2 - 6m + 13 = 0 \qquad m = 3 \pm 2i$$

$$y_{G.S.} = e^{3x}(c_1 \sin 2x + c_2 \cos 2x)$$

$$\boldsymbol{y''' + 3y'' + 3y' + y = 0}$$

$$m^3 + 3m^2 + 3m + 1 = 0 \qquad m = -1, -1, -1$$

$$y_{G.S.} = c_1 e^{-x} + c_2 x e^{-x} + c_2 x^2 e^{-x}$$

$$\boldsymbol{y'''' + 3y'' + 2y = 0}$$

$$m^4 + 3m^2 + 2 = 0 \qquad m = \pm i, \pm\sqrt{2}i$$

$$y_{G.S.} = c_1 \, sin\, x + c_2 \, cos\, x + c_3 \, sin\, \sqrt{2}x + c_4 \, cos\, \sqrt{2}\, x$$

$$\boldsymbol{y^{(3)} + y = 0}$$

$$m^3 + 1 = 0 \qquad m = -1, \frac{1}{2} \pm \frac{\sqrt{3}}{2} i$$

$$y_{G.S.} = c_1 e^{-x} + e^{\frac{x}{2}} \left(c_2 \sin\left(\frac{\sqrt{3}}{2} x\right) + c_3 \cos\left(\frac{\sqrt{3}}{2} x\right)\right)$$

$$\boldsymbol{y^{(4)} + 3y'' - 4y = 0}$$

$$m^4 + 3m^2 - 4 = 0 \qquad m = \pm 1, \pm 2i$$

$$y_{G.S.} = c_1 e^{x} + c_2 \, e^{-x} + c_3 \sin 2x + c_4 \cos 2\, x$$

Example Solve the following DE

$$\boldsymbol{y^{(5)} + 2y^{(3)} + y' = 0}$$

A.E.

$$(m^5 + 2m^3 + m) = 0$$

Therefore, $m(m^4 + 2m^2 + 1) = 0$

$$m(m^2 + 1)^2 = 0$$

Then the values of m are given by $0, \pm i, \pm i$ (repeated)

$$\boldsymbol{y_{G.S.} = c_1 + c_2 sinx + c_3 cosx + c_4 xsinx + c_5 xcosx}$$

Exercise

$$\boldsymbol{y^{(4)} + 2y'' + y = 0\ ,\ y''' + 6y'' = 0}$$

Higher Order Non Homogeneous

$$a_n y^{(n)} + \cdots \ldots + a_2 y'' + a_1 y' + a_0 y = g(x)$$

The solution can be obtained by solving the Homogeneous part alone and get $y_c(x)$ then solve for the Non-Homogenous part and get $y_p(x)$ …. Then the General Solution is $y_{G.S.} = y_c(x) + y_p(x)$

For y_c

Linear DEs Homogeneous with Constant Coefficients

For y_p (depends on $g(x)$)

- ➔ **Linear DEs using Undetermined Coefficients**
- ➔ **Variation of parameters**

HOMOGENEOUS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

Given

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Then, the function $\boldsymbol{y = e^{mx}}$ is a solution of the above DE under n values of m if $a_n \neq 0$

Summary: – $ay'' + by' + cy = 0$

The auxiliary equation is given by:

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Then $m_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ ➔ Three cases :

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Undetermined Coefficients

(y_p for Linear Higher Order Non Homogeneous Constant Coefficients)

Given Linear higher-order DE with constant coefficients in the homogeneous part:-

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + \cdots \ldots . + a_2 y'' + a_1 y' + a_0 y = g(x)$$

where the function $g(x)$ and all its derivatives can be connected in a finite set $\{f_i(x), i = 1,2, \ldots \ldots n\}$. Then, we can assume the particular solution as

$$y_p(x) = \sum_{i=1}^{n} a_i \, f_i(x)$$

Then by substituting in the DE you can get al the coefficients a_i, and hence $y_p(x)$. Examples of $g(x)$:- polynomials, exponential, $\sin(ax)$, $\cos(\mathrm{ax})$, and their products.

Find the set of functions $f_i(x)$ for the following $g(x)$

$g(x)$	$\{f_i(x)\}$
3	a
$x+2$	$ax+b$
$2x^3+2x$	ax^3+bx^2+cx+d
$\sin(2x)$ or $\cos(2x)$	$a\sin(2x)+bcos(2x)$
$2e^{4x}$	ae^{4x}
$(2x+1)sin(2x)$	$(ax+b)\sin(2x)+(cx+d)\cos(2x)$
$(x^2-3x+1)e^{-x}$	$(ax^2+bx+c)e^{-x}$
$e^{3x}\cos(2x)$	$ae^{3x}\sin(2x)+be^{3x}\cos(2x)$
$xe^x\cos(2x)$	$(ax+b)e^x\sin(2x)+(cx+d)e^x\cos(2x)$
$(2x-1)(cosx-e^{2x})$	$(c_1x+d_1)sinx+(c_2x+d_2)cosx+(c_3x+d_3)e^{2x}$
$e^{2x}-\cos(3x)$	$ae^{2x}+bcos(3x)+csin(3x)$
$xcos(2x)+x^2e^{-x}$	$(ax+b)\cos(2x)+(cx+d)\sin(2x)$ $+(a_1x^2+b_1x+c_1)e^{-x}$

Example: $y'' - y' - 2y = 4 - 2e^{5x}$

Solution: $\boldsymbol{y_{G.S.} = y_c + y_p}$

$y_c = c_1 e^{-x} + c_2 e^{2x}$, $y_p = A + Be^{5x}$

Substitute y_p in the DE:

$$(25Be^{5x} - 5Be^{5x} - 2A - 2Be^{5x}) = 4 - 2e^{5x}$$

Comparing coefficients in both sides:

$$18B = -2\,, \qquad -2A = 4$$

$$y_p = -2 - \frac{1}{9}e^{5x}$$

$$\boldsymbol{y_{G.S.} = y_c + y_p}$$

$$y_{G.S.} = c_1 e^{-x} + c_2 e^{2x} - 2 - \frac{1}{9}e^{5x}$$

Example: $y'' + y = 3\sin 2x$

Solution: $\boldsymbol{y_{G.S.} = y_c + y_p}$

$$y_c = c_1 \sin x + c_2 \cos x$$

$$y_p = A\sin 2x + B\cos 2x$$

Substitute in the DE:

$$(-4A\sin 2x - 4B\cos 2x) + (A\sin 2x + B\cos 2x) = 3\sin 2x$$

Comparing coefficients: $A = -1\,, B = 0$

$$y_p = -\sin 2x\,,\ y_{G.S.} = y_c + y_p$$

$$y_{G.S.} = c_1 \sin x + c_2 \cos x - \sin 2x$$

Example: $y'' + y = 3\cos x$

Solution: $y_c = c_1 \sin x + c_2 \cos x$

$y_p = A\, x\, sin\, x + B\, x \cos x$ (Why?!!)

Substitute and compare coefficients:

$$A = \frac{3}{2} \qquad B = 0$$

$$y_{G.S.} = c_1 \sin x + c_2 \cos x + \frac{3}{2} x sin\, x$$

Variation of parameters method

Conside the DE

$$y'' + P(x)y' + Q(x)y = f(x)$$

In case $f(x)$ is not in the domain of the undetermined coefficients method we may use the variation of parameters method as follows:

Given the complementary solution

$$\boldsymbol{y_c(x) = c_1 y_1(x) + c_2 y_2(x)}$$

, then the particular solution takes the form

$$\boldsymbol{y_p(x) = u_1(x) y_1(x) + u_2(x)\, y_2(x)}$$

Variation of parameters

The functions $u_1(x) and\ u_2(x)$ can be obtained from the solution of

$$y_1 u_1' + y_2 u_2' = 0$$

$$y_1' u_1' + y_2' u_2' = f(x)$$

Therefore, $u_1' = \frac{W_1}{W} = -\frac{y_2 f(x)}{W}, and\ u_2' = \frac{W_2}{W} = \frac{y_1 f(x)}{W}$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}, W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}, W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$$

Variation of parameters method (Proof)

$$y'' + p(x)y' + Q(x)y = f(x)$$

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x),$$

$$y_p' = u_1y_1' + y_1u_1' + u_2y_2' + y_2u_2'$$

$$y_p'' = u_1y_1'' + y_1'u_1' + y_1u_1'' + u_1'y_1' + u_2y_2'' + y_2'u_2' + y_2u_2'' + u_2'y_2'.$$

$$\begin{aligned} y_p'' + P(x)y_p' + Q(x)y_p &= u_1[y_1'' + Py_1' + Qy_1] + u_2[y_2'' + Py_2' + Qy_2] + y_1u_1'' + u_1'y_1' \\ &\quad + y_2u_2'' + u_2'y_2' + P[y_1u_1' + y_2u_2'] + y_1'u_1' + y_2'u_2' \\ &= \frac{d}{dx}[y_1u_1'] + \frac{d}{dx}[y_2u_2'] + P[y_1u_1' + y_2u_2'] + y_1'u_1' + y_2'u_2' \\ &= \frac{d}{dx}[y_1u_1' + y_2u_2'] + P[y_1u_1' + y_2u_2'] + y_1'u_1' + y_2'u_2' = f(x). \end{aligned}$$

Imposing the condition:

$$\boldsymbol{y_1 u_1' + y_2 u_2' = 0} \qquad (1)$$

Then we have

$$\boldsymbol{y_1' u_1' + y_2' u_2' = f(x)} \qquad (2)$$

From (1) and (2) , we can determine u_1 and u_2 as given in the formula:

$$u_1' = \frac{W_1}{W} = -\frac{y_2 f(x)}{W} \qquad \text{and} \qquad u_2' = \frac{W_2}{W} = \frac{y_1 f(x)}{W}$$

Variation of parameters (Summary)

$$y'' + P(x)y' + Q(x)y = f(x)$$

First :- Get the complementary solution

$$y_c(x) = c_1 y_1(x) + c_2 y_2(x)$$

Second:- Assume that

$$y_p = u_1(x)y_1(x) + u_2(x)y_2(x)$$

The functions $u_1(x) and\ u_2(x)$ can be obtained from the solution of $y_1 u_1' + y_2 u_2' = 0$ and $y_1' u_1' + y_2' u_2' = f(x)$

Therefore, $u_1' = \frac{W_1}{W} = -\frac{y_2 f(x)}{W}, and\ u_2' = \frac{W_2}{W} = \frac{y_1 f(x)}{W}$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}, W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}, W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$$

Exercise : Solve $y'' + y = \tan x$

Solution: $y_{G.S.} = y_c + y_p$

$$y_c = c_1 \sin x + c_2 \cos x$$

$$y_p = u \sin x + v \cos x$$

$$u' \sin x + v' \cos x = 0$$

$$u' \cos x - v' \sin x = \tan x$$

$$W = \begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix} = -1$$

$$W_1 = \begin{vmatrix} 0 & \cos x \\ \tan x & -\sin x \end{vmatrix} = -\sin x$$

$$W_2 = \begin{vmatrix} \sin x & 0 \\ \cos x & \tan x \end{vmatrix} = \sin x \tan x$$

$$u' = \frac{W_1}{W} = \sin x \quad \rightarrow \quad u = -\cos x$$

$$v' = \frac{W_2}{W} = -\sin x \tan x$$

$$v = -\int \frac{\sin^2 x}{\cos x}\,dx = \int \frac{\cos^2 x - 1}{\cos x}\,dx$$

$$= \int (\cos x - \sec x)dx = \sin x - \ln(\sec x + \tan x)$$

$$y_p = u \sin x + v \cos x$$

$$y_p = -\sin x \cos x + \cos x\,(\sin x - \ln(\sec x + \tan x))$$

$$y_{G.S.} = y_c + y_p$$

Example: $y'' + y = \sec x$

Solution:

$$y_{G.S.} = y_c + y_p$$

$$y_c = c_1 \sin x + c_2 \cos x$$

$$y_p = u(x)\sin x + v(x)\cos x$$

$$u' \sin x + v' \cos x = 0$$

$$u' \cos x - v' \sin x = \sec x$$

$$W = \begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix} = -1 \qquad W_1 = \begin{vmatrix} 0 & \cos x \\ \sec x & -\sin x \end{vmatrix} = -1$$

$$W_2 = \begin{vmatrix} \sin x & 0 \\ \cos x & \sec x \end{vmatrix} = \tan x \qquad u' = \frac{W_1}{W} = 1 \quad \rightarrow \quad u = x$$

$$v' = \frac{W_2}{W} = -\tan x \quad \rightarrow \quad v = \ln|\cos x|$$

$$y_p = x\sin x + \cos x \ln|\cos x|$$

$$y_{G.S.} = c_1 \sin x + c_2 \cos x + x\sin x + \cos x \ln|\cos x|$$

Exercise: $y'' - y = \frac{1}{1+e^x}$

Ans.

$$y = c_1 e^x + c_2 e^{-x} - \frac{1}{2}\left(1 - e^x ln(1 + e^{-x}) + e^{-x} ln(1 + e^x)\right)$$

Example: Solve $4y'' + 36y = csc(3x)$

For y_c: $4y_c'' + 36y_c = 0\,,\ y_{c1} = e^{mx},\ \left(m^2 + 9\right)e^{mx} = 0$

$m^2 + 9 = 0,\ \ m = \pm 3i,\ \ y_c = c_1 sin(3x) + c_2 cos(3x)$

For y_p: (VP method) $\quad y_p = u_1(x)y_1(x) + u_2(x)y_2(x)$

$$W = \begin{vmatrix} sin3x & cos3x \\ 3cos3x & -3sin3x \end{vmatrix} = -3$$

$$W_1 = \begin{vmatrix} 0 & cos3x \\ 0.25csc(3x) & -3sin3x \end{vmatrix} = -0.25\cot(3x)$$

$$W_2 = \begin{vmatrix} sin3x & 0 \\ 3cos3x & 0.25csc(3x) \end{vmatrix} = 0.25$$

$$u_1' = \frac{W_1}{W} = \frac{\cot(3x)}{12} \Rightarrow u_1 = \frac{\ln(sin3x)}{36}$$

$$u_2' = \frac{W_2}{W} = -\frac{0.25}{3} \Rightarrow u_2 = -\frac{x}{12}$$

$$y_p = sin3x\frac{ln(sin3x)}{36} - \frac{x}{12}cos3x$$

$$y_{G.S.} = c_1 sin(3x) + c_2 cos(3x) + sin3x\frac{ln(sin3x)}{36} + -\frac{x}{12}cos3x$$

V. O. P. method for higher order ODE

For third order D.E.

$$y''' + a_2y'' + a_1y' + a_0y = G(x)$$

$$y_c = c_1y_1(x) + c_2y_2(x) + c_3y_3(x)$$

$$y_p = u_1y_1(x) + u_2y_2(x) + u_3y_3(x)$$

$$u_1'y_1 + u_2'y_2 + u_3'y_3 = 0$$

$$u_1'y_1' + u_2'y_2' + u_3'y_3' = 0$$

$$u_1'y_1'' + u_2'y_2'' + u_3'y_3'' = G(x)$$

V. O. P. method for higher order ODE

$$W = \begin{vmatrix} y_1 & y_2 & y_3 \\ y_1' & y_2' & y_3' \\ y_1'' & y_2'' & y_3'' \end{vmatrix} \quad , \quad W_1 = \begin{vmatrix} 0 & y_2 & y_3 \\ 0 & y_2' & y_3' \\ G(x) & y_2'' & y_3'' \end{vmatrix}$$

$$W_2 = \begin{vmatrix} y_1 & 0 & y_3 \\ y_1' & 0 & y_3' \\ y_1'' & G(x) & y_3'' \end{vmatrix} \quad , \quad W_3 = \begin{vmatrix} y_1 & y_2 & 0 \\ y_1' & y_2' & 0 \\ y_1'' & y_2'' & G(x) \end{vmatrix}$$

$$u_1' = \frac{W_1}{W} \quad , u_2' = \frac{W_2}{W} \quad , u_3' = \frac{W_3}{W}$$

V. O. P. method for higher order ODE

Example : Solve the D.E. $\boldsymbol{y''' + y' = \sec x}$

Ans: $\boldsymbol{y_c = c_1 + c_2 \, sin\, x + c_3 \, cos\, x}$

$$\boldsymbol{y_p = u_1(1) + u_2 \, sin\, x + u_3 \, cos\, x}$$

$$u_1' \, (1) + \; u_2' \sin x + u_3' \cos x = \, 0$$

$$u_1'(0) + u_2' \cos x + u_3'(-\sin x) = \, 0$$

$$u_1'(0) + u_2'(-\sin x) + u_3'(-\cos x) = \sec x$$

V. O. P. method for higher order ODE

$$W=\begin{vmatrix}1 & \sin x & \cos x\\ 0 & \cos x & -\sin x\\ 0 & -\sin x & -\cos x\end{vmatrix}=-1,\; W_1=\begin{vmatrix}0 & \sin x & \cos x\\ 0 & \cos x & -\sin x\\ \sec x & -\sin x & -\cos x\end{vmatrix}=-\sec x$$

$$W_2=\begin{vmatrix}1 & 0 & \cos x\\ 0 & 0 & -\sin x\\ 0 & \sec x & -\cos x\end{vmatrix}=\tan x \quad , \quad W_3=\begin{vmatrix}1 & \sin x & 0\\ 0 & \cos x & 0\\ 0 & -\sin x & \sec x\end{vmatrix}=1$$

$$u_1'=\frac{W_1}{W}=\sec x \quad \rightarrow \quad u_1=\ln(\sec x+\tan x)$$

$$u_2'=\frac{W_2}{W}=-\tan x \quad \rightarrow \quad u_2=\ln(\cos x)$$

$$u_3'=\frac{W_3}{W}=-1 \quad \rightarrow \quad u_3=-x$$

$$\boldsymbol{y_p=\ln(\sec x+\tan x)+\ln(\cos x)\sin x-x\cos x}$$

$$\boldsymbol{y_{G.S.}=y_c+y_p}$$

Higher Order (Variable Coeffocients)

Cauchy Eular Equation: (General form)

$$(\boldsymbol{ax}+\boldsymbol{b})^{\boldsymbol{n}}\boldsymbol{a_n y^{(n)}}+\cdots+(\boldsymbol{ax}+\boldsymbol{b})^2\boldsymbol{a_2 y''}+(\boldsymbol{ax}+\boldsymbol{b})\boldsymbol{a_1 y'}+\boldsymbol{a_0 y}=\boldsymbol{G}(\boldsymbol{x})$$

Is called Cauchy Eular equation

Let $ax+b=e^t \to t=\ln(ax+b)$

$$y'=\frac{dy}{dx}=\frac{dy}{dt}\frac{dt}{dx}=\frac{a}{ax+b}\frac{dy}{dt} \quad \Rightarrow \ (ax+b)y'=a\dot{y}$$

$$(ax+b)y''+ay'=a\frac{d\dot{y}}{dx}=a\frac{d\dot{y}}{dt}\frac{dt}{dx}=\frac{a^2}{ax+b}\ddot{y}$$

$$(ax+b)^2y''=a^2\,(\ddot{y}-\dot{y})$$

We can show that : $(ax+b)^3y'''=a^3\,(\dddot{y}-3\ddot{y}+2\dot{y})$

The D.E. is transformed to constant coefficients one in y, t

Higher Order (Variable Coeffocients)

Cauchy Eular Equation: (Spcial case)

If $\boldsymbol{a = 1}$ and $\boldsymbol{b = 0}$

$$\boldsymbol{a_n x^n y^{(n)} + \cdots + a_2 x^2 y'' + a_1 x y' + a_0 y = G(x)}$$

Let $x = e^t \rightarrow t = \ln x$

$$xy' = \dot{y}$$

$$x^2 y'' = (\ddot{y} - \dot{y})$$

$$x^3 y''' = (\dddot{y} - 3\ddot{y} + 2\dot{y})$$

Summary: Cauchy Euler Des

$$a_2x^2y'' + a_1xy' + a_0y = g(x) \qquad (1)$$

let $x = e^t$, $\dot{y} = \frac{dy}{dt}$, $\ddot{y} = \frac{d^2y}{dt^2}$, then

$$xy' = \dot{y}\,, \quad x^2y'' = \ddot{y} - \dot{y} \quad , \quad x^3y''' = \dddot{y} - 3\ddot{y} + 2\dot{y}$$

Then, (1) becomes

$$a_2(\ddot{y} - \dot{y}) + a_1\dot{y} + a_0y = g(e^t)$$

A.E. of y_c: $\quad a_2\left(m^2 - m\right) + a_1m + a_0 = 0$

Or, $\quad a_2m(m-1) + a_1m + a_0 = 0$

$$a_3x^3y''' + a_2x^2y'' + a_1xy' + a_0y = g(x) \qquad (2)$$

A.E. of y_c:

$$a_3m(m-1)(m-2) + a_2m(m-1) + a_1m + a_0 = 0$$

Higher Order (Variable Coeffocients)

Cauchy Eular Equation:

Solve the D. E

$$(\mathbf{2}\boldsymbol{x}+\mathbf{1})^{\mathbf{2}}\boldsymbol{y}''+\mathbf{8}(\mathbf{2}\boldsymbol{x}+\mathbf{1})\boldsymbol{y}'+\mathbf{8}\boldsymbol{y}=\mathbf{8}(\mathbf{2}\boldsymbol{x}+\mathbf{1})^{\mathbf{2}}$$

Let $2x+1=e^{t}\rightarrow t=\ln(2x+1)$

$(2x+1)y'=2\dot{y}$

$(2x+1)^{2}y''=2^{2}\,(\ddot{y}-\dot{y})$

$2^{2}\,(\ddot{y}-\dot{y})+8(2\dot{y})+8y=8e^{2t}$ => $4\ddot{y}+12\dot{y}+8y=8e^{2t}$

$\ddot{y}+3\dot{y}+2y=2e^{2t}$ => Aux. eqn $m^{2}+3m+2=0$

$m=-1,-2 \qquad y_{c}=c_{1}e^{-t}+c_{2}e^{-2t}=c_{1}\frac{1}{(2x+1)}+c_{2}\frac{1}{(2x+1)^{2}}$

Let $y_{p}=Ae^{2t} \qquad \dot{y}_{p}=2Ae^{2t} \qquad \ddot{y}_{p}=4Ae^{2t}$

$4Ae^{2t}+6Ae^{2t}+2Ae^{2t}=2e^{2t}$ => $A=\frac{1}{6}\ y_{p}=\frac{1}{6}e^{2t}=\frac{1}{6}x^{2}$

Example:- Solve the following DE $x^2y'' - xy' + y = lnx$

First you should identify that it is Cauchy Euler problem:-

$x = e^t, t = lnx$ then the DE will be $(\ddot{y} - \dot{y}) - \dot{y} + y = t$

$$\ddot{y} - 2\dot{y} + y = t$$

For $y_c(t)$ ➔

$$\ddot{y}_c - 2\dot{y}_c + y_c = 0$$

Assume $y_c(t) = e^{mt}$

$$m^2 - 2m + 1 = 0$$

$$m = 1,1$$

$$y_c(t) = c_1e^t + c_2te^t$$

$$y_c(x) = c_1x + c_2xln(x)$$

For $y_p(t)$ ➔ UC

$$\ddot{y}_p - 2\dot{y}_p + y_p = t$$

Assume $y_p(t) = a + bt$

$$\dot{y}_p = b, \qquad \ddot{y}_p = 0$$

$$L.H.S. = 0 - 2b + a + bt = t$$

$$b = 1, a = 2$$

$$y_p(t) = 2 + t$$

$$y_p(x) = 2 + \ln(x)$$

$$\boldsymbol{y_{G.S.} = c_1x + c_2xln(x) + 2 + ln(x)}$$

Example:- Solve the following DE:

a) $x^2y'' - 2xy' - 4y = 0$

$$\boldsymbol{x = e^t, t = lnx, (\ddot{y} - \dot{y}) - 2\dot{y} - 4y = 0 \Rightarrow \ddot{y} - 3\dot{y} - 4y = 0, y_c = e^{mt} \Rightarrow m^2 - 3m - 4 = 0 \Rightarrow m = 4, -1 \Rightarrow y(t) = c_1e^{4t} + c_2e^{-t} \Rightarrow y(x) = c_1x^4 + \frac{c_2}{x}}$$

b) $4x^2y'' + 8xy' + y = 0$

$$\boldsymbol{x = e^t, t = lnx\,, y_c = e^{mt} \Rightarrow \quad 4m(m-1) + 8m + 1 = 0}$$

$$\boldsymbol{\Rightarrow 4m^2 + 4m + 1 = 0 \Rightarrow m = -2, -2 \Rightarrow y(t) = c_1e^{-2t} + c_2te^{-2t}}$$

$$\boldsymbol{\Rightarrow y(x) = c_1x^{-2} + c_2x^{-2}ln(x)}$$

c) $x^3y''' + 5x^2y'' + 7xy' + 8y = 0$

$$\boldsymbol{x = e^t, t = lnx\,, y_c = e^{mt} \Rightarrow}$$

$$\boldsymbol{m(m-1)(m-2) + 5m(m-1) + 7m + 8 = 0}$$

$$\boldsymbol{\Rightarrow m^3 + 2m^2 + 4m + 8 = 0 \Rightarrow m = -2, \pm 2i}$$

$$\boldsymbol{\Rightarrow y(t) = c_1e^{-2t} + c_2sin(2t) + c_3cos(2t)}$$

$$\boldsymbol{\Rightarrow y(x) = c_1x^{-2} + c_2sin(2lnx) + c_3cos(2lnx)}$$

Example:- Solve the DE: $x^2y'' + xy' - y = \frac{1}{x+1}$

$x = e^t, t = \ln x$ then the DE will be $(\ddot{y} - \dot{y}) + \dot{y} - y = \frac{1}{1+e^t}$

$$\ddot{y} - y = \frac{1}{1 + e^t}$$

For $y_c(t)$➔ $\ddot{y}_c - y_c = 0$ Assume $y_c(t) = e^{mt} \Rightarrow m^2 - 1 = 0$

$\Rightarrow m = -1,1$ ➔ $y_c(t) = c_1e^{-t} + c_2e^t$ ➔ $y_c(x) = \frac{c_1}{x} + c_2x$

For $y_p(t)$➔ VP (Let us work in x) $y'' + \frac{y'}{x} - \frac{y}{x^2} = \frac{1}{x^2(x+1)}$

$y_c(x) = \frac{c_1}{x} + c_2x,\ y_1(x) = \frac{1}{x}, y_2(x) = x,\ W = \begin{vmatrix} 1/x & x \\ -1/x^2 & 1 \end{vmatrix} = \frac{2}{x}$

$W_1 = \begin{vmatrix} 0 & x \\ \frac{1}{x^2(x+1)} & 1 \end{vmatrix} = -\frac{1}{x(x+1)}$ ➔ $u_1' = \frac{-1}{2(x+1)}$ ➔ $u_1 = -0.5\ln(x+1)$

$W_2 = \begin{vmatrix} 1/x & 0 \\ -1/x^2 & \frac{1}{x^2(x+1)} \end{vmatrix} = \frac{1}{x^3(x+1)}$ ➔ $u_2' = \frac{1}{2x^2(x+1)} = \frac{1}{2x^2} - \frac{1}{2x} + \frac{1}{2(x+1)}$

$\Rightarrow u_2 = -\frac{1}{2x} - 0.5\ln x + 0.5\ln(x+1)$

$$\boldsymbol{y_{G.S.} = \frac{c_1}{x} + c_2x - \frac{0.5\ln(x+1)}{x} - 0.5 - 0.5x\ln x + 0.5x\ln(x+1)}$$

Example: $2x^2y''' + 6xy'' + 2y' - 2(y/x) = 1$

Solution: Multiply both sides by x

$2x^3y''' + 6x^2y'' + 2xy' - 2y = x$ **(Cauchy Euler Form)**

$$2(\dddot{y} - 3\ddot{y} + 2\dot{y}) + 6(\ddot{y} - \dot{y}) + 2\dot{y} - 2y = e^t$$

$$2\dddot{y} - 2y = e^t$$

$$y_c\text{: } A.E. \quad 2m^3 - 2 = 0 \qquad m = 1, -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

$$y_c = c_1e^t + e^{-\frac{1}{2}t}(c_2 sin\frac{\sqrt{3}}{2}t + c_3 cos\frac{\sqrt{3}}{2}t)$$

$$y_p = Ate^t \qquad \dot{y}_p = A(1+t)e^t$$

$$\ddot{y}_p = A(2+t)e^t \qquad \dddot{y}_p = A(3+t)e^t$$

$$2A(3+t)e^t - 2Ate^t = e^t \qquad 6A = 1 \quad A = 1/6$$

$$y_p = \frac{1}{6}te^t$$

$$y_{G.S.}(t) = c_1 e^t + e^{-\frac{1}{2}t}\left(c_2 \sin\frac{\sqrt{3}}{2}t + c_3 \cos\frac{\sqrt{3}}{2}t\right) + \frac{1}{6}te^t$$

$$y_{G.S.}(x) = c_1 x + \left(1/\sqrt{x}\right)\left(c_2 \sin\frac{\sqrt{3}}{2}(\ln x) + c_3 \cos\frac{\sqrt{3}}{2}(\ln x)\right) + \frac{1}{6}x \ln x$$

Reduction of Order method (R.O.)

a) Homogenuous D.E.:

Consider the Second Order homogenuous D.E.

$$a_2(x)\, y'' + a_1(x)\, y' + a_0(x)\, y = 0$$

Given y_1 is a solution.

The general solution $y = c_1 y_1 + c_2 y_2$

y_2 can be obtained by using Abel's formula

$$y_2 = y_1 \int \frac{1}{y_1^2} e^{-\int \frac{a_1(x)}{a_2(x)} dx}\, dx$$

Reduction of Order method (R.O.)

Proof of Abel's formula (reduction of order)

Let $y_2 = uy_1$

$$y_2' = u'y_1 + uy_1'$$

$$y_2'' = u''y_1 + 2u'y_1' + uy_1''$$

Substitute in D.E.

$$a_2(x)[u''y_1 + 2u'y_1' + uy_1''] + a_1(x)[u'y_1 + uy_1'] + a_0(x)uy_1 = 0$$

$$a_2(x)u''y_1 + u'[2a_2(x)y_1' + a_1(x)y_1] + u\underbrace{[a_2(x)y_1'' + a_1(x)y_1' + a_0(x)y_1]}_{0} = 0$$

Let $u' = T \rightarrow u'' = T'$ **in last equation and divide by** $a_2(x)y_1$

$$T' + \left(\frac{2y_1'}{y_1} + \frac{a_1(x)}{a_2(x)}\right)T = 0 \text{ (linear or separable d.e.)}$$

$$\frac{dT}{T} = -\left(\frac{2y_1'}{y_1} + \frac{a_1(x)}{a_2(x)}\right)dx \rightarrow \ ln\, T = -2ln\, y_1 - \int \frac{a_1}{a_2}dx$$

$$T = \frac{1}{y_1^2}e^{-\int \frac{a_1(x)}{a_2(x)}dx} \qquad \rightarrow \qquad u = \int T dx = \int \frac{1}{y_1^2}e^{-\int \frac{a_1(x)}{a_2(x)}dx}dx$$

Reduction of Order method (R.O.)

Proof of Abel's formula (reduction of order)

$$y_2 = uy_1$$

$$\boxed{y_2 = y_1 \int \frac{1}{y_1^2} e^{-\int \frac{a_1(x)}{a_2(x)} dx} \, dx}$$

This is called the Abel's formula

***Abel's formula** finds the second part y_2 of the homogenuous D.E. when y_1 is given or known.*

The steps of R.O. Method in proof can be applied to solve the Nonhomogenuous D.E. given y_1 is a solution of the homogenuous part.

Reduction of Order method (R.O.)

Example 1:

Solve the D.E. $(x^2+1)y''-2x\,y'+2y=0$ if $y_1=x$ is a solution to the D.E.

Solution:

$$y_2 = y_1 \int \frac{1}{{y_1}^2} e^{-\int \frac{a_1(x)}{a_0(x)}dx}\, dx$$

$$y_2 = x \int \frac{1}{x^2} e^{\int \frac{2x}{x^2+1}dx}\, dx$$

$$y_2 = x \int \frac{1}{x^2} e^{\ln(x^2+1)}\, dx = x \int \frac{1}{x^2}(x^2+1)dx$$

$$y_2 = x \int \left(1+\frac{1}{x^2}\right) dx = x\left(x-\frac{1}{x}\right)$$

$$y_2 = (x^2-1)$$

$$y = c_1\, x + c_2(x^2-1)$$

Reduction of Order method (R.O.)

Example 2:

Solve the D.E. $y'' - 3y' + 2y = 0$ if $y_1 = e^x$ is a solution to the D.E.

Solution:

$$y_2 = y_1 \int \frac{1}{{y_1}^2} e^{-\int \frac{a_1(x)}{a_0(x)} dx} \, dx$$

$$y_2 = e^x \int \frac{1}{e^{2x}} e^{\int 3\,dx} \, dx$$

$$y_2 = e^x \int \frac{1}{e^{2x}} e^{3x} \, dx$$

$$y_2 = e^x \int e^x dx$$

$$y_2 = e^x e^x = e^{2x}$$
$$y = c_1 e^x + c_2 e^{2x}$$

Reduction of Order method (R.O.)

Example 3:

Solve the D.E. $(2x+1)y''-4(x+1)\ y'+4y=0$ if $y_1 = e^{2x}$ is a solution to the D.E.

Solution:

$$y_2 = y_1 \int \frac{1}{{y_1}^2} e^{-\int \frac{a_1(x)}{a_0(x)} dx}\ dx$$

$$y_2 = e^{2x} \int \frac{1}{e^{4x}} e^{\int \frac{4(x+1)}{2x+1} dx}\ dx$$

$$y_2 = e^{2x} \int \frac{1}{e^{4x}} e^{\int \left(2+\frac{2}{2x+1}\right) dx}\ dx = e^{2x} \int \frac{1}{e^{4x}} e^{2x+\ln(2x+1)}\ dx$$

$$y_2 = e^{2x} \int \frac{1}{e^{4x}} (2x+1) e^{2x} dx$$

$$y_2 = e^{2x} \int (2x+1) e^{-2x} dx$$

$$y_2 = e^{2x} * (-xe^{-2x} - e^{-2x}) = -x-1$$

$$y_2 = -x-1$$

$$y = c_1 e^{2x} + c_2(-x-1) = c_1 e^{2x} + c_3(x+1)$$

Reduction of Order method (R.O.)

Example 4:

Solve the D.E. $x^2y'' - 6x\,y' + 10y = 3x^4 + 6x^3$ if $y_1 = x^2$ is a solution to the homogenous part of the D.E.

Solution:

$y = y_1 v = x^2 v$
$y' = 2xv + x^2v'$
$y'' = 2v + 2xv' + 2xv' + x^2v''$
$y'' = 2v + 4xv' + x^2v''$

Sub in the D.E.

$x^2(2v + 4xv' + x^2v'') - 6x(2xv + x^2v') + 10x^2v = 3x^4 + 6x^3$
$x^4v'' - 2x^3v' = 3x^4 + 6x^3$

Let $T = v'$
$T' = v''$
$x^4T' - 2x^3T = 3x^4 + 6x^3$

Reduction of Order method (R.O.)

$T' - \frac{2}{x}T = 3 + \frac{6}{x}$ (first order linear D.E.)

$\mu = e^{\int -\frac{2}{x}dx} = e^{-2\ln x}$

$\mu = \dfrac{1}{x^2}$

$\dfrac{1}{x^2}T = \int \dfrac{1}{x^2}\left(3 + \dfrac{6}{x}\right)dx$

$\dfrac{1}{x^2}T = -\dfrac{3}{x} - \dfrac{3}{x^2} + c_1$

$T = -3x - 3 + c_1x^2$

$v = \int (-3x - 3 + c_1x^2)\,dx$

$v = -\dfrac{3}{2}x^2 - 3x + \dfrac{c_1}{3}x^3 + c_2$

$y = x^2v = -\dfrac{3}{2}x^4 - 3x^3 + \dfrac{c_1}{3}x^5 + c_2x^2$

$y_h = \frac{c_1}{3}x^5 + c_2x^2$ $y_p = -\frac{3}{2}x^4 - 3x^3$

Reduction of Order method (R.O.)

Example 5:

Solve the D.E. $x^2y'' - x(x+2)\ y' + (x+2)y = x^3$ if $y_1 = x$ is a solution to the homogenous part of the D.E.

Solution:

$y = y_1 v = xv$
$y' = v + xv'$
$y'' = v' + v' + xv''$
$y'' = 2v' + xv''$

Sub in the D.E.

$x^2(2v' + xv'') - x(x+2)(v + xv') + (x+2)xv = x^3$
$2x^2v' + x^3v'' - x^2(x+2)v' = x^3$
$x^3v'' - x^3v' = x^3$
$v'' - v' = 1$

Let $T = v'$
$T' = v''$
$T' - T = 1$

Reduction of Order method (R.O.)

$T' - T = 1$ (first order linear D.E.)

$\mu = e^{\int -1 dx}$

$\mu = e^{-x}$

$e^{-x}T = \int e^{-x}(1)dx$

$e^{-x}T = -e^{-x} + c_1$

$T = -1 + c_1 e^{x}$

$v = \int (-1 + c_1 e^{x})dx$

$v = -x + c_1 e^{x} + c_2$

$y = xv = -x^2 + c_1 x e^{x} + c_2 x$

$y(1) = 1 \quad \therefore -1 + c_1 e^{1} + c_2 * 1 = 1$

$y'(1) = 2 \quad \therefore -2(1) + c_1(e^{1} + 1e^{1}) + c_2 = 2$

$c_1 = 2e^{-1}, c_2 = 0$

$y = -x^2 + 2e^{-1} x e^{x}$

Reduction of Order method (R.O.)

Example 6:

Solve the D.E. $\sin^2 x\, y'' - 2\sin x \cos x\; y' + (\cos^2 x + 1)y = \sin^3 x$ if $y_1 = \sin x$ is a solution to the homogenous part of the D.E.

Solution:

$y = y_1 v = \sin x\, v$
$y' = \cos x\, v + \sin x\, v'$
$y'' = -\sin x\, v + \cos x\, v' + \cos x\, v' + \sin x\, v''$
$y'' = -\sin x\, v + 2\cos x\, v' + \sin x\, v''$

Sub in the D.E.

$\sin^2 x(-\sin x\, v + 2\cos x\, v' + \sin x\, v'') - 2\sin x \cos x\,(\cos x\, v + \sin x\, v') + (\cos^2 x + 1)\sin x\, v = \sin^3 x$
$v'' \sin^3 x = \sin^3 x$
$v'' = 1$
$v' = x + c_1$
$v = \dfrac{x^2}{2} + c_1 x + c_2$
$y = \sin x\, v$
$y = \left(\dfrac{x^2}{2} + c_1 x + c_2\right)\sin x$

Reduction of Order method (R.O.)

Example: Use R.O method to get particular solution of the D.E. :

$$2xy'' + xy' - y = 2x^3e^{-\frac{x}{2}}$$

Given $y_1 = x$ is one of its homogeneous equation solutions.

Answer: Let $y = ux$, $y' = u'x + u$, $y'' = u''x + 2u'$

$$2x(u''x + 2u') + x(u'x + u) - ux = 2x^3e^{-\frac{x}{2}}$$

$$2x^2u'' + u'\left(4x + x^2\right) = 2x^3e^{-\frac{x}{2}}$$

$$u'' + u'\left(\frac{2}{x} + \frac{1}{2}\right) = xe^{-\frac{x}{2}}$$

Let $u' = z$, $u'' = z'$

Reduction of Order method (R.O.)

$\therefore\ z' + z\left(\frac{2}{x} + \frac{1}{2}\right) = xe^{-\frac{x}{2}}$ **=> Linear 1st order D.E.**

$$\mu = e^{\int \frac{2}{x}+\frac{1}{2}dx} = e^{2ln(x)+\frac{x}{2}} = x^2 e^{\frac{x}{2}}$$

$$z = \frac{1}{x^2 e^{\frac{x}{2}}}\left[\int x^2 e^{\frac{x}{2}} x e^{-\frac{x}{2}} dx + c_1\right] = \frac{1}{x^2 e^{\frac{x}{2}}}\left[\frac{x^2}{4} + c_1\right]$$

$$z = \frac{x^2}{4} e^{-\frac{x}{2}} + c_1 \frac{e^{-\frac{x}{2}}}{x^2}$$

$$u = \int \frac{x^2}{4} e^{-\frac{x}{2}} dx + c_1 \int \frac{e^{-\frac{x}{2}}}{x^2} dx + c_2$$

$$y_p = x \int \frac{x^2}{4} e^{-\frac{x}{2}} dx = x\left[-2e^{-\frac{x}{2}}\left(\frac{x^2}{4} + 2\left(\frac{2x}{4}\right)\right) + 4\left(\frac{2}{4}\right)\right] = e^{-\frac{x}{2}}\left(-\frac{x^3}{2} - 2x^2 - 4x\right)$$

Example: Solve the DE

$$y'' + 2y' \tan x + (1 + 2tan^2 x)y = 0$$

Given that $y = cos\ x$ is a solution

Solution: $y_{G.S.} = y_1 u = u \cos x$

$$u(x) = c_1 \int \frac{e^{-\int P(x)dx}}{y_1^2} dx + c_2$$

$$e^{-\int p(x)dx} = e^{-\int 2\tan x\ dx} = e^{2\ln\cos x} = cos^2 x$$

$$u(x) = c_1 \int \frac{cos^2 x}{cos^2 x} dx + c_2 = c_1 x + c_2$$

$$y_{G.S.} = u \cos x = c_1 x cos\ x + c_2 \cos x$$

End of Lecture